Project Management Plan

SQRT (Sub-zero Quantitative Research Team)

Orlaith Ni Dhuill, Elyana Abdallah, Javier Ayuso, Bruce Ritter, Caimin Keavney

Document History

Date	Version	Notes	Approver
2026–02–25	SQT-CDR-PMP-0	Initial release	CK

Author Contributions

OND §1–7

EA §4

JA §6

BR §1–6

CK §2, Reviewer

Contents

1 Executive Summary **3**

2 Project Overview **4**
2.1 Mission Objectives 4

3 Organisational Structure **5**
3.1 Key Responsibilities 5

4 Work Breakdown Structure **6**
4.1 Work Packages 7

5 Project Timeline **9**
5.1 Mission Phases 10

6 Budget & Procurement **11**
6.1 Initial Budget 11
6.2 Bill of Materials 11
6.3 Procurement Plan 11

7 Risk Register **12**

List of Figures

1 SQRT Mission Organigram 5
2 SQRT Mission Work Breakdown Structure 6
3 SQRT Mission Gannt Chart 9
4 Risk Register Matrix of SQRT Mission 12

List of Tables

1 SQRT Key Responsibilities 5
2 SQRT Work Packages 7
3 Key Mission Dates 9
4 Bill of Materials 11
5 Risk Register 12

Key Reference Documents

1. SQT-PDR-PMP-0
2. SQT-CDR-SEP-0
3. SQT-PDR-MDD-0
4. UCD-SCD-2026-0

1 Executive Summary

This document presents our management plan for the SQRT mission, including the organisational and work breakdown structure, the project timeline, the mission budget, procurement schedule and the associated risk register. Key technical risks are discussed, along with mitigation strategies. The project management plan outlined in this document will facilitate the completion of the mission goals and requirements in a timely and cost-effective manner.

2 Project Overview

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to investigate how such plumes thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes and water desalinisation processes will be done using a high-altitude demonstration. SQRT will provide a controlled and feasible approach to studying evaporation-driven thermal behaviour in a low-pressure environment which could be useful for future studies relating planetary and icy-moon plume activity [3],[1],[4],[5], or for future terrestrial applications relating desalination processes [6],[8],[7] for drinking water. Multistage flash desalination is a growing sector in water desalination and the SQRT mission aims to complete similar processes using this background research as guidance in our design. The design of the project includes a mechanism to expose liquid water to another container kept close to the temperature and pressure of the outside near-space environment at a set altitude. The temperature of the water sample will be measured overtime via a thermal sensor as it undergoes a predicted flash. A more detailed description of the mission plan can be found in SQT-PDR-MDD-0.

The project management plan is affected by the mission objectives, mission requirements (Level 1), and technical requirements (Level 2), which are outlined in SQT-PDR-MDD-0 and SQT-CDR-SEP-0. These requirements affect the overall mission design. Additionally, the budget and procurement strategy is affected by the component choices in the design overview in SQT-CDR-HDF-0. The scheduling of work packages is influenced by the procurement strategy, the Level 1 and Level 2 requirements, and key milestones outlined in Sec. 5.

2.1 Mission Objectives

The project management plan aims to introduce the timely completion of the following SQRT mission objectives:

- **MO1** Fly and recover a TupperSat containing an experimental scientific or engineering/technology demonstration payload with potential applications to space science.

- **MO2** SQRT will attempt to reproduce and observe the flash evaporation process of water exposed to near-space conditions.

3 Organisational Structure

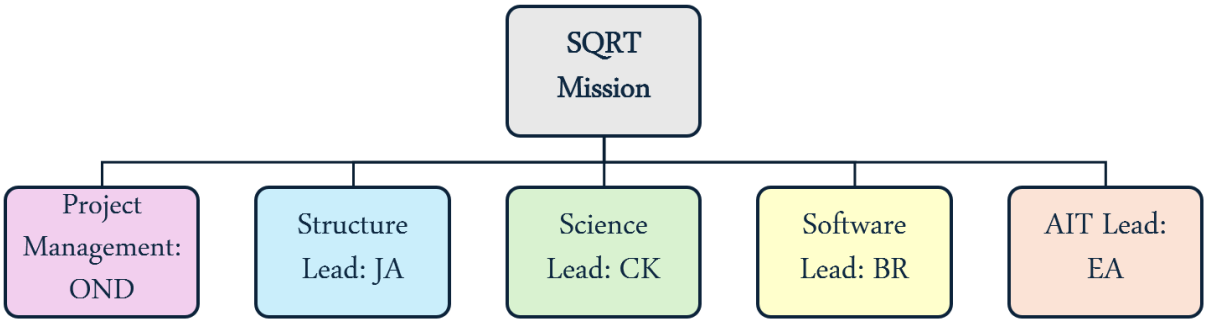


Figure 1: SQRT Mission Organigram

The SQRT project is overseen by five lead engineers: a project manager, a science lead, a structure lead, a software lead, and an AIV lead. These lead engineers will oversee key parts of the SQRT project, based on their background. These sections correspond to five of the main work packages in the work breakdown structure of this mission.

However, the SQRT team will collaborate on all aspects of the mission so team leads will not hold full responsibility over all sub work packages within the section they lead. Responsibility for individual work packages are detailed within Section 4.

3.1 Key Responsibilities

In addition to work packages, team members are responsible for authoring and reviewing mission documents. The key responsibilities and authored documents are listed by team member name in Table 1 below.

Table 1: SQRT Key Responsibilities

Team Member	Key Responsibilities	Document Authorship
Elyana Abdallah	Leading AIV	SQT-CDR-VP and SQR-CDR-FTSTP Author, other documents TBA
Javier Ayuso Vazquez	Leading Structure Development	SQT-CDR-HDF and SQT-PDR-DJF Author, other documents TBA
Caimin Keavney	Leading Science	SQT-CDR-SEP, SQT-CDR-DMP and SQT-PDR-SEP Author, other documents TBA
Orlaith Ni Dhuill	Project Manager	SQT-CDR-PMP and SQT-PDR-MDD Author, other documents TBA
Bruce Ritter	Leading Software Development	SQT-CDR-SDF and SQT-PDR-PMP Author, other documents TBA

4 Work Breakdown Structure

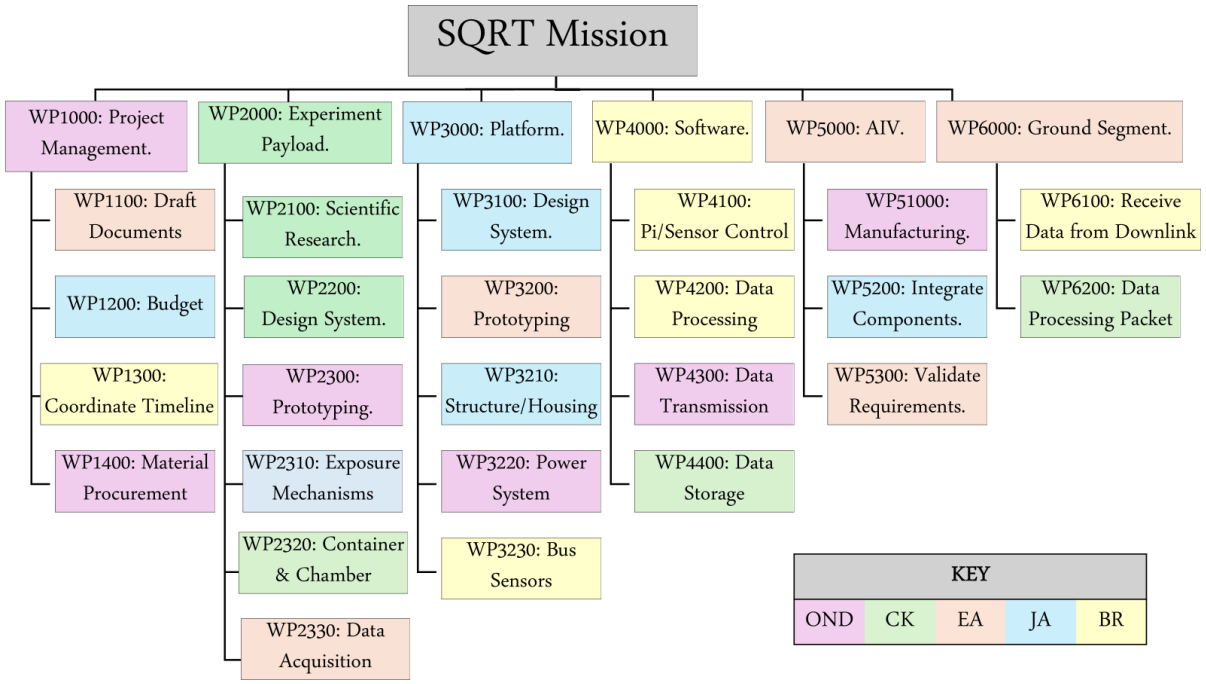


Figure 2: SQRT Mission Work Breakdown Structure

The SQRT mission will follow the work breakdown structure shown in Figure 2 above to ensure that all necessary work for the SQRT mission is completed within the schedule and all requirements detailed in SQT-CDR-SEP-0 are met.

While work on SQRT systems is collaborative, individual team members are assigned the final responsibility of ensuring completion of individual work packages.

Responsibility for signing off on individual work packages is assigned by colour using the key in the lower right-hand corner of Figure 2.

Leadership roles of subsections of the SQRT mission are assigned in Section following the same colour coding rubric. These subsections align with work packages WP1000, WP2000, WP3000, WP4000, and WP5000.

4.1 Work Packages

The following work packages found in Table 2 define the work that will be done to complete the SQRT mission requirements and achieve mission goals. Work packages are also represented on a Gantt chart timeline in Section 5.

Table 2: SQRT Work Packages

WP Lvl	WP Title	Task	Prerequisites & Dependencies	Deliverables	Start	End	Owner
1	WP1000	Complete the SQRT Project Management	WP1100, WP1200, WP1300, WP1400	PDR, CDR, FAR management deliverables	20/1	28/4	OND
2	WP1100	Draft Mission Documents	Mission concept, requirements	PDR, CDR, FAR Documentation	20/1	12/4	EA
2	WP1200	Define and Track Project Budget	BOM, Procurement Plan	Approved and tracked budget	20/1	21/2	JA
2	WP1300	Coordinate Project Timeline	WBS, review milestones	Integrated project schedule	20/1	28/4	BR
2	WP1400	Procure Project Components	BOM, Budget approval	Procured hardware and materials	25/2	28/2	OND
1	WP2000	Complete Experiment payload	WP2100, WP2200, WP2300, WP5000	Functional experiment payload	19/1	3/4	CK
2	WP2100	Perform Science Research	Literature, mission goals	Scientific justification and models	19/1	27/2	CK
2	WP2200	Design Experiment Payload	Requirements, science results	Payload design ideas	26/1	27/2	CK
2	WP2300	Prototype Experiment Payload	WP2200	Payload prototype, completed WP2310, WP2320, WP2330.	24/2	6/3	OND
3	WP2310	Develop Exposure Mechanism	Payload design	Functional exposure mechanism	24/2	6/3	JA
3	WP2320	Develop Container and Chamber	Structural and thermal requirements	Container and chamber	24/2	6/3	CK
3	WP2330	Prototype Payload Data Acquisition System	Sensor requirements	Experiment data acquisition system	24/2	6/3	EA
1	WP3000	Develop Platform Subsystem	WP3100, WP3200, WP5000	Functional platform subsystem	26/1	3/4	JA
2	WP3100	Design Platform Architecture	System requirements	Platform design documentation, initial payload design ideas	2/2	27/2	JA

2	WP3200	Prototype platform	WP3100	Platform prototype, completed WP3310, WP3320, WP3330.	24/2	6/3	EA
3	WP3210	Prototype Structure and Housing	Structural design	Platform structure and housing	24/2	6/3	JA
3	WP3220	Create Power System	Power requirements	Operational power subsystem	24/2	19/3	OND
3	WP3230	Prototype Platform Sensors	Sensor specifications	Integrated platform sensors	9/3	19/3	BR
1	WP4000	Software	System requirements	Flight software suite	16/2	3/4	BR
2	WP4100	Develop Onboard Control Software	Pi/Sensor interfaces	Pi/Sensor control software	16/2	3/4	BR
2	WP4200	Develop Data Processing Software	Raw experiment data	Processed data products	9/3	3/4	BR
2	WP4300	Develop Data transmission software	Communications requirements	Data transmission software	9/3	3/4	OND
2	WP4400	Develop Data Storage Software	Storage Requirements	Onboard data storage system	16/2	9/3	CK
1	WP5000	Perform Assembly, Integration and Verification	WP5100, WP5200, WP5300	Integrated and verified system ready for FAR	9/3	3/4	EA
2	WP5100	Manufacture System Components	Final designs, WP2200, WP3200.	Manufactured Components	9/3	19/3	OND
2	WP5200	Integrate Subsystems	Completed subsystems, WP5100.	Integrated TupperSat	16/3	3/4	JA
2	WP5300	Validate Requirements via Testing	Integrated system, WP5200, WP4000.	Verified requirements	16/3	3/4	EA
1	WP6000	Operate Ground Segment	FAR complete, payload launch.	Ground segment operations WP6100, WP6200, WP6300.	14/4	25/4	EA
2	WP6100	Receive Data from Downlink	Receiver	Received flight data	14/4	25/4	BR
2	WP6200	Process Downlinked Data	Raw telemetry	Processed flight data	14/4	25/4	CK

5 Project Timeline

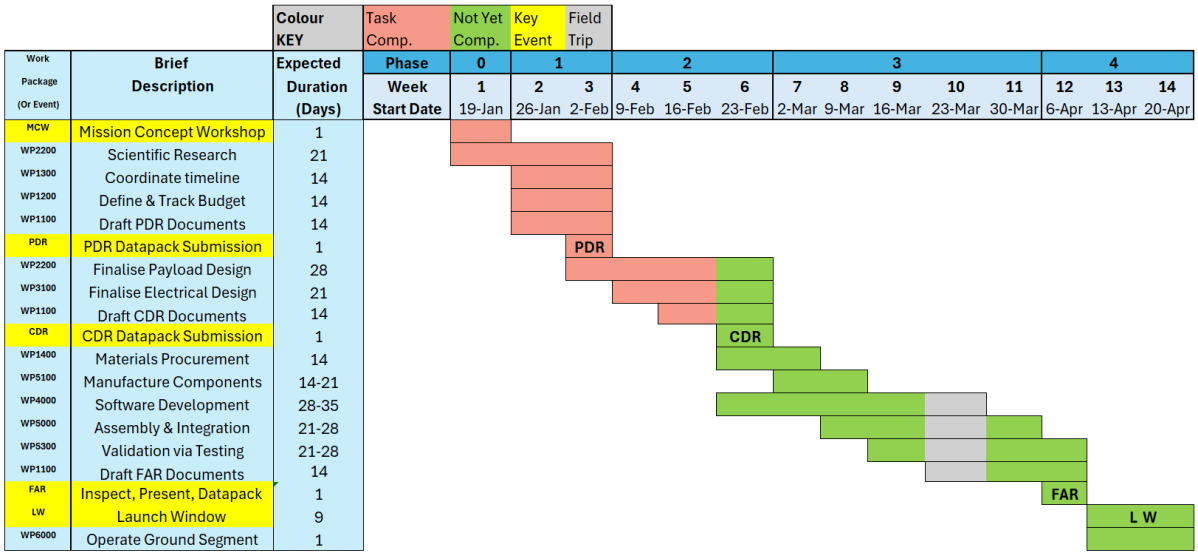


Figure 3: SQRT Mission Gannt Chart

The schedule our team will follow to develop, manufacture, test, and launch the SQRT mission is shown in the updated Gantt chart above, which details the full mission scope from the Mission concept workshop (MCW) all the way through to the Mission close-out review (MCR). The mission work packages are implemented clearly along with a brief description and their expected duration assigned to their relative phase and week expectancy of completion. The key events are highlighted clearly in yellow. These milestones and other key dates have been pulled directly from UCD-SCD-2026-0, and are described in Table *3*. Certain tasks can be started before earlier tasks are finished, such as prototyping which will start with in-house components before procurement is done but cannot be fully completed until after all the necessary products have been acquired. The total duration of this mission will be 99 days from the kick-off meeting on January 19th, 2026 until the feedback session with UCD on April 28th, 2026.

Table 3: Key Mission Dates

Dates (2026)	**Event**	**Deliverables**	**Description**
19/01	KOM	None	Project Kick Off
06/02	PDR	PDR Data Pack	Interview with UCD
27/02	CDR	CDR Data Pack	Interview with UCD
10/04	FAR	TupperSat	Presentation to UCD
14-25/04	Launch Window	TupperSat	11 Day duration Launch Window
28/04	Mission Review & Feedback Session	Mission Analysis	Meeting to review objective completion
19/05	MCR Submission	MCR	Individual Analysis

5.1 Mission Phases

Phase 0: The mission concept and goals are developed following the kick-off meeting.

Phase 1: An initial system design and project requirements are drafted, and the PDR documents are written. This phase ends with the Preliminary Design Review (PDR).

Phase 2: Following acceptance of the PDR, the design is matured and prototyping and software development work begins. Materials are procured and CDR documents are written. This phase ends with the Critical Design Review (CDR).

Phase 3: Following acceptance of the CDR, procurement, prototyping, and software development steps are completed. The final TupperSat design is manufactured and tested, including long range tests. This phase ends with the Flight Acceptance Review (FAR).

Phase 4: The SQRT TupperSat is launched and operates.

Phase 5: Data from the mission operation is analysed and the mission is reviewed.

6 Budget & Procurement

6.1 Initial Budget

Funding from University College Dublin for this mission is set to €100.00 with a contingency of €20.00 (total capped amount: €120). All necessary materials not provided by UCD will be procured using this funding. A list of the materials to be procured is shown in the bill of materials in Table 4 The budget for this mission is calculated based on the expected costs of items (and shipping and VAT where necessary). This list of materials is derived from the design overview in SQT-CDR-HDF-0. Items ordered from UCD suppliers Mouser and Farnell will not incur shipping costs for this mission. There are no labor costs associated with this project as all team members are unpaid students. The current budgeted expenses total €92.71, leaving a margin of €7.29 (excluding the additional €20 contingency).

6.2 Bill of Materials

Table 4: Bill of Materials

Item	**Descriptor**	**Quantity**	**Source**	**Cost (€)**
Adafruit MLX90640 Thermal Sensor	Thermal sensor array to observe thermal evolution of flash-exposure event during experiment.	1	Mouser, Supplier ID: 485-4469	Unit price; 67.04. With VAT; 82.46
Water source	Distilled water will be used to reduce uncontrolled variables in the experiment.	500ml	School of Chemistry, UCD.	0.00
Tupperware Container; 'Clip n Close Lunchbox'	A Tupperware contained maximum 2L in volume will be used to encapsulate the SQRT payload. It will not be structural.	1	Aldi	4.99 (VAT Incl.)
Servo Motor	Used in pinch valve mechanism for experiment payload.	1	Farnell Part No. SER0043	Unit price; 4.24. With VAT; 5.26
5ml Syringe	A syringe will be modified and used as the water container.	2	School of Medicine UCD	0.00
Medical Tubing	Used in the pinch valve.	Approx 10cm	School of Medicine UCD	0.00
			Total Cost incl. VAT	€92.71

6.3 Procurement Plan

Procurement will be completed by the start of week 7. The SQRT timeline has 2 weeks set for procurement to accommodate order processing and shipping. The procurement of soft materials such as tubing, syringes, Tupperware and insulation bubble-wrap will be completed directly by the project manager. Electrical components such as the thermal array sensor and servo motor will be ordered through the school. Some materials will be 3D printed and team manufactured such as the valve mechanism and the chamber dividers and most electrical materials will be provided by the School of Physics Laboratory.

7 Risk Register

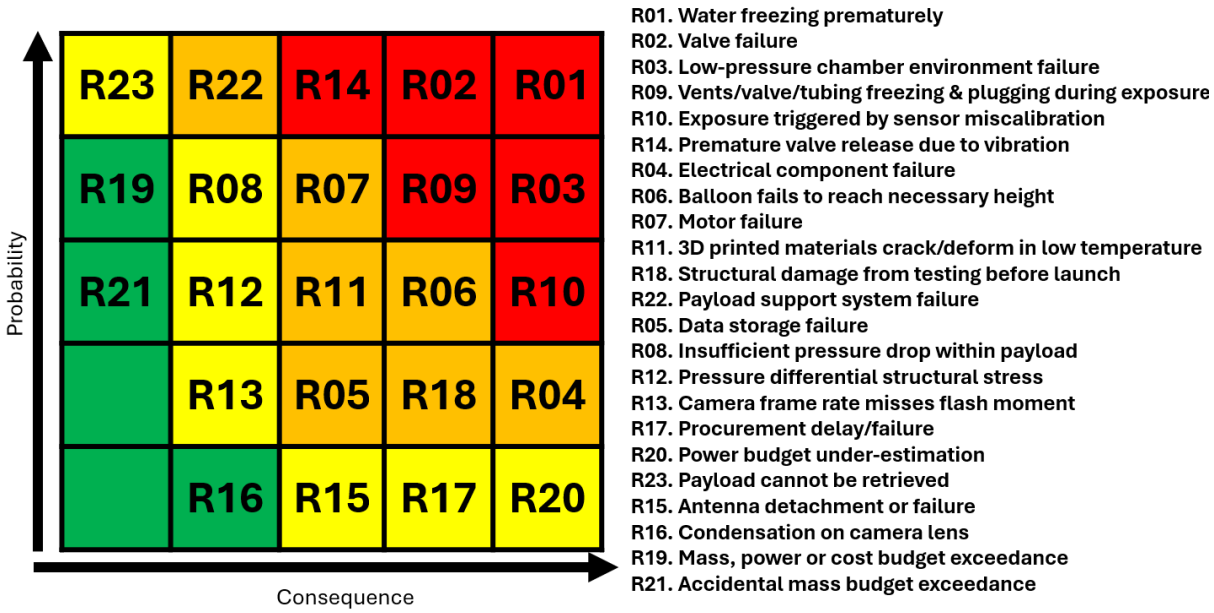


Figure 4: Risk Register Matrix of SQRT Mission

The Risk Register identifies and tracks potential threats to the SQRT mission throughout the design and implementation phases. Each risk is assessed in terms of its probability (P, on a scale of A-E) of occurrence and the severity of impact on mission success (I, on a scale of 1-5). These factors are qualitatively classified in risk levels using a Low (L), Medium (M), or High (H) scale.

The overall risk level is determined by combining the probability and impact assessments. Risks assessed as High will be prioritized for mitigation. Active mitigation measures will be taken prior to flight approval. Medium risks are monitored and reduced where feasible. Low risks are considered acceptable but are periodically reviewed. These mitigation strategies are designed to reduce either the likelihood of risk occurrence, the severity of impact, or both to ensure that the residual risk remains within acceptable project limits. A list of risks to the SQRT mission are evaluated in Table 5.

Table 5: Risk Register

ID	Risk	Description	P	I	Risk Level	Mitigation
R01	Water freezing prematurely	Water freezes within chamber before release exposure due to low external temperatures of -40°C to -60°C	E	5	H	Sufficient insulation of chamber and tubing to valve exposure location using bubble-wrap or other thermal insulators in addition to dissipated heat source provided by the electrical operations of the TupperSat.

R02	Valve failure	The release valve fails to operate, jams or releases prematurely resulting in breakage, potentially caused by software or mechanical fault	E	4	H	Performance of repeated and measurable testing and verification of valve operation and reliability.
R03	Low-pressure chamber environment failure	Venting of low-pressure chamber is not sufficient and prevents the exposure chamber from reaching an external ambient pressure environment suitable for the flash-evaporation to occur	D	5	H	Conduct ground based pressure testing and verification on the low-pressure chamber to ensure its robustness.
R04	Any or all electrical component failure	Some or all electrical components fail during flight, with potential causes due to water leakage, power disturbance, temperature drop, software issues, signal-loss, mechanical problems	B	5	M	Ensure all electrical components are stored safely from water chamber, without interference, with proper insulation and separation from low pressure chamber, ensure sufficient on the ground validation and performance testing to ensure reliable operation during flight.
R05	Data Storage Failure	SD Card failure or power interruption which could result in any or complete loss of attained data from the flight experiment	B	3	M	Ensure periodic writing and verification, transmit only essential relevant data telemetry during the flight to ensure SD Card is unlikely to fail or undergo interruption, repeated on the ground verification and testing
R06	Balloon fails to reach necessary height	The balloon bursts prematurely to the expected altitude due to environmental conditions, weakness of structural material, lack of sufficient helium. Therefore resulting in trigger condition not being met	C	4	M	Have fail-safe procedures in place for rapid loss of altitude whereby the valve releases prematurely in the event of premature balloon burst. Lower mission expectations of the trigger altitude and ensure software is allowable for such a risk.
R07	Motor failure	The Servo motor fails to perform the desired mechanical movement to release the pinch valve due to software or mechanical failure.	D	3	M	Repeated ground-based testing and verification to ensure reliability.

R08	Insufficient pressure drop within payload	The pressure drop from the pressurised liquid chamber through the tubing to the exposure chamber is not enough to create a flash evaporation	D	2	L	Ensure the non low-pressure elements of the TupperSat are sufficiently sealed and designed in a way that the pressure drop is substantial.
R09	Venting, Valve or tubing rapid freezing and plugging during exposure event	This could occur if the low pressure and temperatures within the exposure chamber freeze instantly during the exposure event, resulting water freeze/plugging before experiment and as such lack of observation, and therefore lack of data acquisition	D	4	H	Testing of exposure event equipment reliability with in sub-zero freezing temperatures.
R10	Exposure event triggered by sensor miscalibration	The pressure/altitude/ temperature sensors miscalibrate the desired trigger conditions resulting in early release of valve, therefore occurring in undesired environmental conditions and obtained results	C	5	H	Repeated ground based and non-ground based testing to ensure accurate calibration and operation of all sensors onboard to reduced probability of misfire caused by miscalibration.
R11	3D Printed materials crack/ deform in low temperature environment	If any unintended exposure occurs to external temperatures the 3D printed materials may potentially break, crack, or become brittle	C	3	M	Use of high durability materials that prove reliability and perform well in ground based testing. Good design to avoid any low temperature exposure.
R12	Pressure difference caused stress to structure	It is possible that the pressure differential between the low-pressure and the remaining apparatus may experience structural stress and damage	C	2	L	Design the structure of the the TupperSat in such a way that the pressure differential is accounted for and perform stress tests to avoid damage.
R13	Camera failure or frame rate misses the flash evaporation moment	It could occur that the exact moment the flash evaporation occurs the camera may be unable to capture it due to misalignment of frame-rate	B	2	L	Design the payload in such a way that the camera is positioned within in a broad field of view and ensure it is capturing data frame at optimal frame/second speeds

R14	Vibration during launch may lead to premature valve release	During launch the environment and movement or unexpected turbulence of the of the TupperSat may cause vibrational stress and and mechanical misalignment within the payload causing the pinch valve to release prematurely	E	3	H	Perform ground based test to find a quantitative lower limit at which the pinch valve may handle vibrational movement and design the payload in a way that this vibrational limit cannot be met.
R15	Antenna detachment or failure	As the antenna is exposed externally and may experience increased motion and rotational movement than that of the payload within the constraints of the Tupperware its possible it may fail or detach from the payload during flight	A	3	L	Design the payload in such a way that the security and operation of the antenna is prioritised and successfully undergoes vibrational and high rotational velocity testing.
R16	Condensation on camera lens/protective cap	As the camera will be positioned to observe the exposure event within the exposed low-pressure and low temperature chamber, it is possible that condensation may form on the camera which could interfere with the obtained thermal evolution data as the datasheet[2] for the MLX90640 explicity states that the field of view must be entirely clear for best results	A	2	L	The design of the payload should account for this ad mitigate accordingly or the condensation shall be considered in post data analysis.
R17	Procurement of materials delay/failure	In the event that any of the materials are not procured or are delayed in procurement, this could delay the whole work-flow of the mission and as such could cause the launch window to be missed.	A	4	L	The materials will shall be procured and tested with amble time window prior to launch. Failure to procure any of the required mission materials shall be dealt with by obtaining the 'next-best' option with haste.
R18	Damage to structural integrity during testing before launch	This mission relies heavily on pre-launch testing and its possible the stuctural integrity of this TupperSat may become damaged during this process	B	4	M	Environmental testing shall be well prepared and shall not be completed in such a way that structural damage can easily occur prior to launch of the TupperSat.

R19	Mass, Power or Cost Budget exceedance	There are strict limitations in terms of mass, power and cost budgets within this mission. Over use/indulgence of any mentioned budgets may result in future problems for the mission and mission documentation	D	1	L	The budgets and limitations of the mission shall be considered at every review checkpoint/stage during the mission.
R20	Power budget underestimation	The payload relies on a set power limitation of of the battery pack, if for any reason electrical bus requires more than that estiamted this coul lead to powerloss of the whole system	A	5	L	Ensure sufficient preparation in power required estimations so the payload power supply shall not be exceeded or underestimated
R21	Accidental mass budget exceedance	This mission has a strict mass budget which could be easily overlooked at later stages of the mission. Small late stage additions like insulation, wiring and mounts could push the mass of the TupperSat beyond its budget and could cause equilibrium problems	C	1	L	The mass budget should be adhered to in such a way to allow contingency in case of late stage material addition
R22	The payload support system fails leading to loss of payload	In the event the payload support system and connection with the balloon fails, the TupperSat may be irretrievable and may not complete the exposure event which would be detrimental to the mission.	E	2	M	The payload shall be designed in such a way that this risk is optimally reduced and shall be vigorously tested in ground-based simulation
R23	The payload cannot be retrieved	In the event the payload cannot be retrieved, the onboard data would be lost and this would be a loss to post-flight analysis	E	1	L	The payload shall be designed in such a way that the onboard GPS and antenna shall be operational and allow for optimal recovery post-flight. Impact is also reduced by transmitting science data during flight

References

[1] Plume activity on europa: Current knowledge and search strategy for europa clipper. *The Planetary Science Journal*, 6(8):182, 2025.

[2] Adafruit. *Adafruit MLX90640 Micropython GitHub*, 2025.

[3] M. K. Dougherty, K. K. Khurana, F. M. Neubauer, C. T. Russell, J. Saur, J. S. Leisner, and M. E. Burton. Identification of a dynamic atmosphere at enceladus with the cassini magnetometer. *Science*, 311(5766):1406–1409, 2006.

[4] C. J. Hansen, D. E. Shemansky, L. W. Esposito, A. I. F. Stewart, B. R. Lewis, J. E. Colwell, A. R. Hendrix, R. A. West, Jr. Waite, J. H., B. Teolis, and B. A. Magee. The composition and structure of the enceladus plume. *Science*, 311(5766):1422–1425, 2006.

[5] Hans L. F. Huybrighs, Yoshifumi Futaana, Stanislav Barabash, Martin Wieser, Peter Wurz, Norbert Krupp, Karl-Heinz Glassmeier, and Bert Vermeersen. On the in-situ detectability of europa's water vapour plumes from a flyby mission. *Icarus*, 285:270–282, 2017.

[6] O. Miyatake, T. Tomimura, Y. Ide, and T. Fujii. An experimental study of spray flash evaporation.

[7] Sami Mutaira and Yasuyuki Ikegami. Experimental study on flash evaporation from superheated water jets: Influencing factors and formulation of correlation.

[8] Karolina Wojtasik. *Boiling of Water at Low Pressure: The Role of Subcooling and Thermophysical Properties on the Bubble Dynamics and Heat Transfer*. PhD thesis, Université de Lyon and Wroclaw University of Science and Technology, 2021. Thèse de doctorat. HAL Id: tel-03433365, version v2.